Introduction to Economics

Class 13 — Monopoly; oligopoly and the prisoner's dilemma

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9 November 2026

Plan for today

- Monopoly: the price and quantity a single seller picks
- What a monopoly costs the rest of us
- A price for each buyer; what governments do
- Oligopoly: two sellers and a deal between them
- More sellers; when a deal can hold
- Practice: today's moves; a tax, a tariff, an external cost

Madrid–Barcelona by fast train: one company, then three

Revealed in class

Before class: Ouigo (May 2021) and Iryo (November 2022) started running rival trains on this line. Guess what had happened by 2025 to the average fare and to the number of passengers — and why.

Source given in class.

From the second hundred on, each adds less to revenue than its price: the price falls for all

highest price for all (euros)	13	12	11	10	9	8	7	6
rides a day	100	200	300	400	500	600	700	800

Marginal revenue: what one more ride adds to revenue
(here: per ride, for each extra hundred)

Monopoly: the only seller of a good with no close substitute; it faces the market's demand (class 12, slide 20).

A price taker's extra bag adds just the price: marginal revenue (MR) = P = 4 (class 12, slides 6 and 8).

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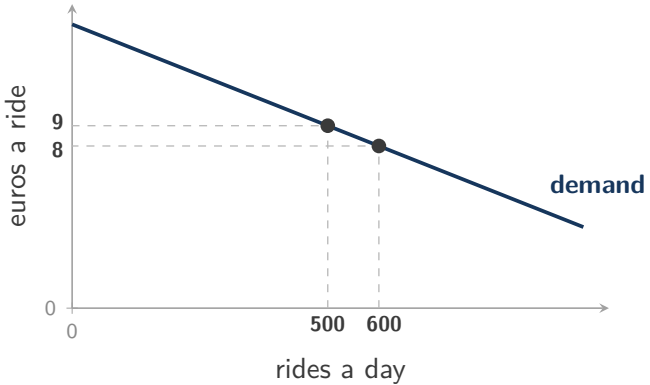
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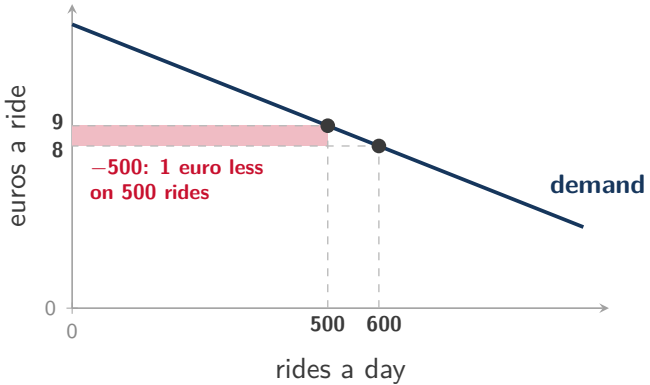
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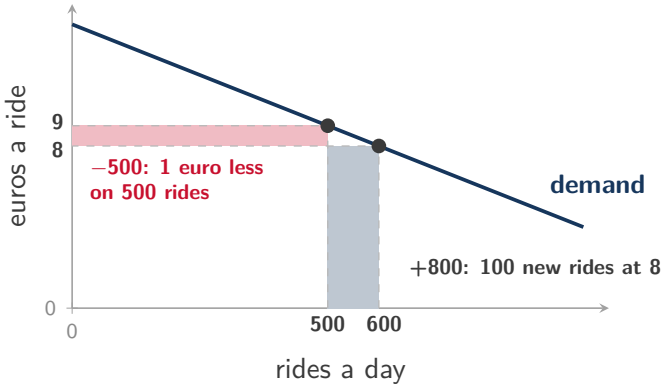
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



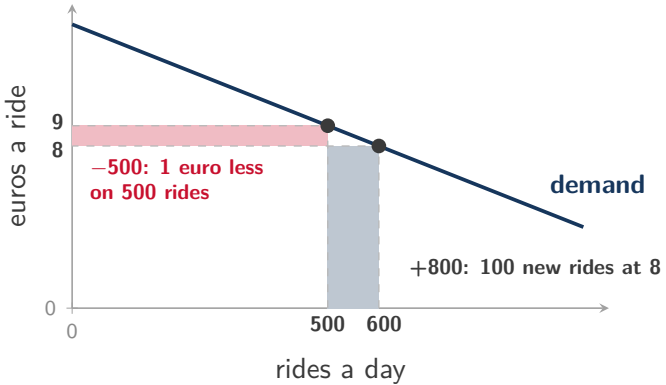
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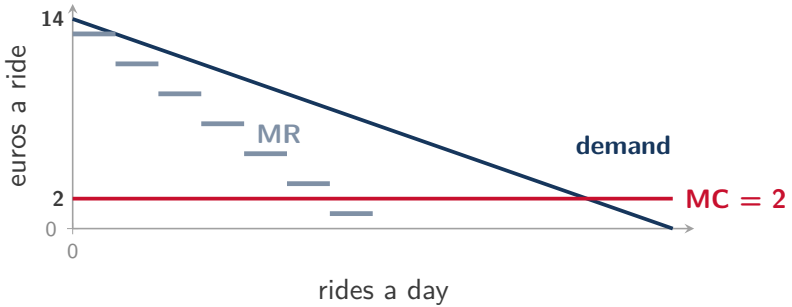
From 500 to 600 rides: 800 more from new riders, 500 less from the old ones



Marginal revenue: $800 - 500 = 300$ for 100 rides — 3 a ride, below the price of 8.

Which rides does the cable car sell?

Each extra ride costs 2 euros (energy, wear); fixed costs are paid whatever it sells.



Before class: which is the last hundred whose marginal revenue beats the 2-euro cost? What price goes with it?

Marginal cost (MC): what one more ride adds to total cost (class 11, slide 10). A price taker produces while MC is below the price (class 12, slide 8).

What does the cable car's choice cost?

Revealed in class

Before class: at the cable car's price, some riders who value a ride at more than its 2-euro cost do not ride. How many rides would be sold at a price of 2? Who loses what? (The demand line: price $= 14 -$ rides $\div 100$.)

A different price for each rider?

Revealed in class

Price discrimination: selling the same good at different prices to different buyers.
Before class: if every rider paid exactly what the ride is worth to them, how many rides would the cable car sell, and who would gain the surplus?

Four answers to a monopoly: competition law, a price cap, public ownership — or nothing

- **Competition law:** stop mergers, fine price-fixing

Natural monopoly: one firm serves the whole market most cheaply (class 12, slide 21).

In Spain: the CNMC (Comisión Nacional de los Mercados y la Competencia). Reading Ch. 11, pp. 258–261.

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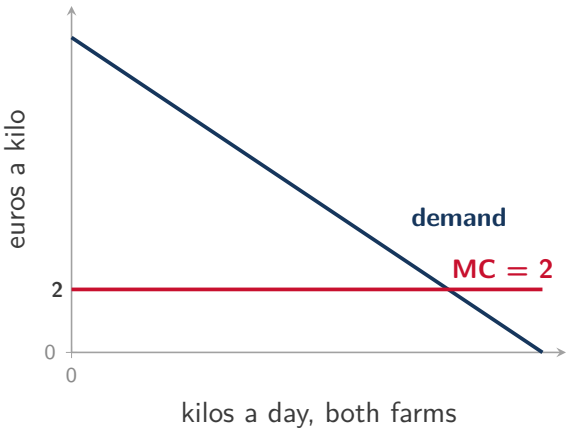
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- **Doing nothing:** when every cure costs more

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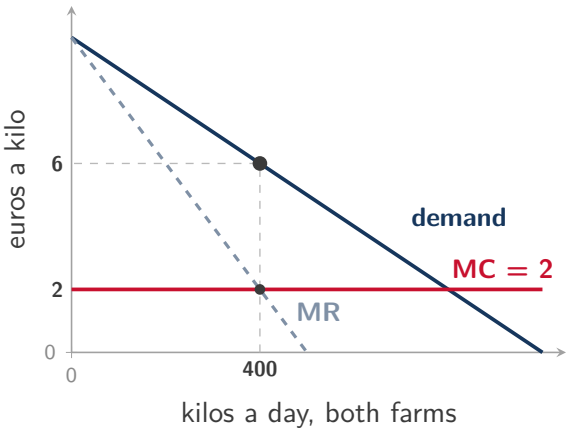
Two mussel farms share one bay: acting as one monopoly, they would sell 400 kilos at 6 euros



Perla and Onada: the only two farms; each kilo costs 2 to farm.

Oligopoly: a market with a few sellers of similar goods

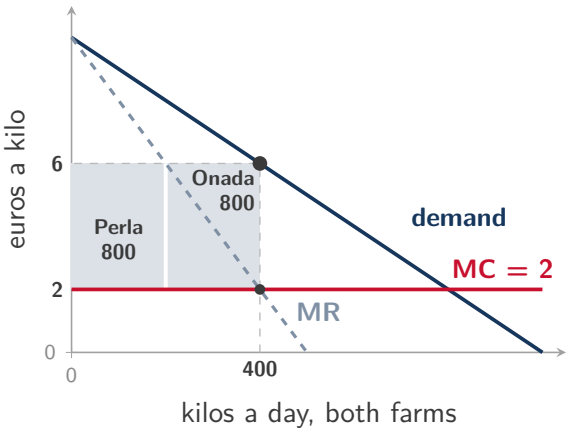
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Cartel: sellers who agree to act as one monopoly: here 200 kilos each

Keep the deal or sell more? Each farm's profit a day

Each farm sells 200 kilos (the deal) or 250.
Price: 400 kilos $\rightarrow$ 6.00 · 450 $\rightarrow$ 5.50 · 500 $\rightarrow$ 5.00.

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

$$\text{profit} = (\text{price} - 2) \times \text{own kilos}$$

Where do the two farms end up?

euros a day, Perla's first

		Onada	
		200 kilos	250 kilos
Perla	200 kilos	Perla 800, Onada 800	Perla 700, Onada 875
	250 kilos	Perla 875, Onada 700	Perla 750, Onada 750

Dominant strategy: best whatever the other does

Nash equilibrium: each does its best, given what the other does

Prisoner's dilemma: each does its best, and both are worse off than if both had cooperated

Revealed in class

Before class: as one monopoly the two farms sell 400 kilos at 6 euros; each kilo costs 2. Where does the price go when each farm does its best for itself, and when many farms share the bay? Why?

Meeting every day: can the deal hold?

Revealed in class

Before class: Perla breaks the deal on day 1, and from day 2 both sell 250 kilos every day. How much does Perla gain on day 1, and lose on each day after? Tit-for-tat: start with the deal, then do what the other did yesterday.

A monopoly sells while marginal revenue beats marginal cost and reads its price off the demand curve.

Its price is above marginal cost: rides worth more than their cost go unsold.

Break

Ten-minute break

Multiple choice, two minutes alone: the monopoly's price

A1 [4 points]

One firm owns the town's only fibre network. Each extra household it connects costs it 10 euros a month. The table gives the highest price at which it can connect each number of households, one price for all. Which price gives the firm the largest profit?

price (euros a month)	60	50	40	30	20	10
households (hundreds)	1	2	3	4	5	6

A 20 euros a month
B 40 euros a month
C 10 euros a month
D 30 euros a month

Answer A1

A B C D

Multiple choice, two minutes alone: open on Sundays?

A2 [4 points]

A town's only two supermarkets, Fresc and Rodó, each choose whether to open on Sundays. Profit in thousand euros a month, Fresc's first:

		Rodó: closed	Rodó: open
Fresc	closed	Fresc 12, Rodó 10	Fresc 8, Rodó 13
	open	Fresc 15, Rodó 6	Fresc 9, Rodó 8

Which outcome is the Nash equilibrium?

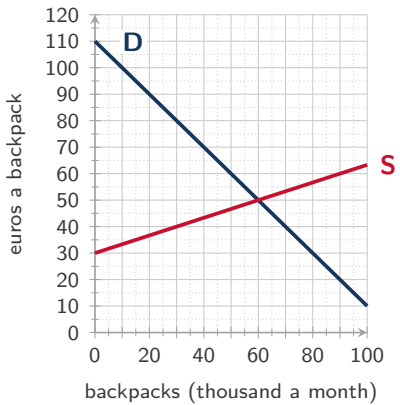
A Fresc: closed; Rodó: closed.
B Fresc: open; Rodó: closed.
C Fresc: closed; Rodó: open.
D Fresc: open; Rodó: open.

Answer A2

A B C D

A tax on backpacks: the line and the prices — three minutes alone

The grid shows the market for backpacks: demand D and supply S. The state puts a tax of 40 euros on every backpack sold; the sellers pay it to the state.



[5 points; drawing 2; boxes as marked]

(a) Draw the supply curve with the tax: at every quantity 40 euros above S. Label it S + tax.

(b) With the tax (quantity in thousand, prices in euros):

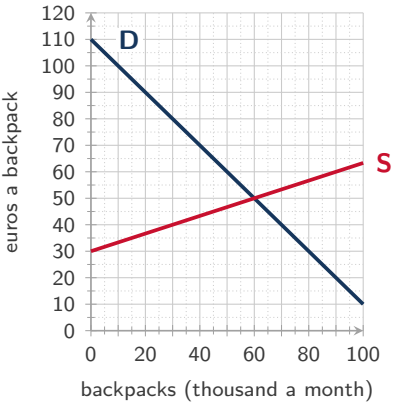
Quantity sold [1]

Buyers pay (euros a backpack) [1]

Sellers keep (euros a backpack) [1]

Draw on the grid and label every line you draw; numbers in the boxes, working beside them.

The tax's areas: numbers, and letters inside — five minutes alone



[9 points; letters 2; boxes as marked]

(c) With the tax of 40 euros a backpack, in thousand euros a month: CS, PS, T and DWL (below). Write each one's letters inside its area.

CS [1.5]	PS [1.5]	T [1.5]	DWL [1.5]

(d) Who bears more of the tax? Circle one. [1]

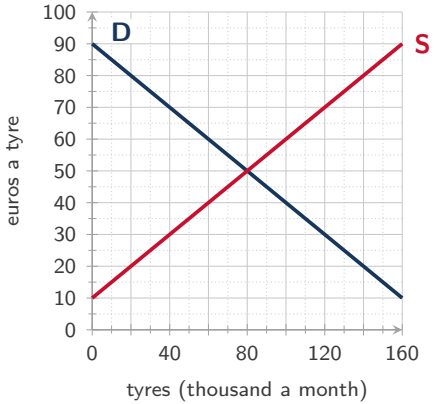
Buyers Sellers

(e) Same tax, but the buyers pay it to the state: which line would you draw? Would any box change?

CS: consumer surplus · PS: producer surplus · T: tax revenue · DWL: deadweight loss. An area's letters go inside it (a small area: beside it, with an arrow). A triangle is ½ × base × height.

Tyres: opening to trade, then a tariff — four minutes alone

The grid shows the market for tyres in a country that does not trade: demand D and supply S.



[6.5 points; drawing 2; boxes as marked]

(a) The country opens to trade at a world price of 30 euros a tyre. Draw it as a line; label it "world".

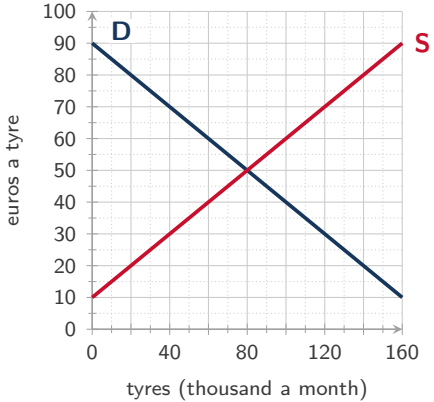
Production [1] Consumption [1] Imports [0.5]

(b) Then a tariff of 10 euros on every imported tyre. Draw the price with the tariff; label it "tariff".

Price [0.5] Production [0.5]

Consumption [0.5] Imports [0.5]

The tariff's areas: T and two DWL triangles — three minutes alone



[5 points; letters 1; boxes as marked]

(c) With the tariff (world price 30, tariff 10), in thousand euros a month: the tariff revenue (T) and the deadweight loss (DWL, both triangles together). Write T inside its area, and DWL beside each of its two triangles, with an arrow.

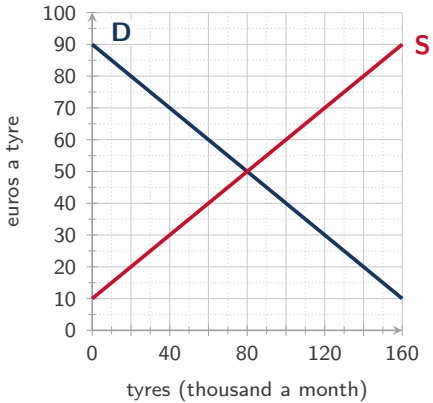
T [2]

DWL (total) [2]

Thousand tyres × euros = thousand euros. A triangle is ½ × base × height.

The same tyres with smoke: an external cost — four minutes alone

The tyre market without trade, D and S as before. Making each tyre also sends out smoke that harms people nearby: 30 euros of harm per tyre, which buyers and sellers do not count.



[10 points; drawing 4; boxes as marked]

(a) Draw the social-cost curve (SC: the sellers' cost plus the harm to others); label it SC.

(b) Market quantity [1] Efficient quantity [1.5]

(c) The deadweight loss (DWL) at the market quantity, thousand euros a month; label its area DWL.

(d) The tax per tyre that brings the market to the efficient quantity.

(c) DWL [2] (d) Tax (euros a tyre) [1.5]

Hoodies: from costs to a choice — four minutes alone (optional)

[8 points; boxes as marked]

A workshop sews hoodies and sells them at the market price. Its total cost, euros a week:

hoodies a week	0	25	50	75	100
total cost (euros)	400	700	900	1,300	1,900
marginal cost, MC (euros a hoodie)					

(a) Fill in the marginal cost (MC) of each extra 25 hoodies, per hoodie. [2]

(b) At 50 hoodies: the average variable cost (AVC)? The average total cost (ATC)? [2]

(c) The price is 20 euros a hoodie. How many hoodies? Profit? [2]

(d) The price falls to 14. Produce or shut down? Why? [2]

Variable cost = total cost − fixed cost (class 11, slides 9–11).
A price taker produces while MC < price (class 12, slide 8).

Before Thursday 12 November

- Thursday 12 November: monopolistic competition; problems on Blocks 3–4 — read Mankiw & Taylor Ch. 12, pp. 267–276
- Problem set 3: paper copy handed out today; bring your worked solutions and your copy on Thursday (not handed in)
- Practise in Sextant: the units the course page lists